MODULE 06 | VERSION 3.0

Construction & Post-Handover

Governance-aligned controlled consolidation | 25 October 2026

DOCUMENT CONTROL	APPLICATION
Status	Harmonized business specification; approved prior module workflow retained with controlling Module 7 alignment.
Composition	This V3 addendum and the full previously revised module source are combined in a single PDF.
Priority	This V3 alignment governs in case of conflict. The original module source remains operative where not contradicted.
Central governance	Module 7 is a continuous operational backbone; it is not a gate to be completed after Module 6.
Authority	Documented GM-led delegation and owner-reserved matters replace generic or inconsistent old “Management alone” labels.
Scope of this module	Preconstruction and mobilization; site workforce; materials, procurement, subcontractors, equipment, safety, quality, variation, costs and post-handover obligations.

1. Implementation interpretation and source preservation

This document intentionally retains the complete existing revised module specification after this amendment. Do not erase earlier workflow, screenshots, forms, stages or user-approved protections. Configure the shared functions from the V3 Master Architecture and Module 7 specification. The addendum is not permission to skip work or approvals.

2. Shared mandatory company controls

- GOV-01 Governance: The owner retains reserved strategic powers and protected oversight, while the employed General Manager (GM) holds company-wide operational accountability. Owner access to reporting does not authorize routine operational interference.
- GOV-02 Delegation: Department heads, Project Managers and designated officers may authorize decisions only within documented project, function, category, value and risk limits; above-limit or exceptional decisions go to GM. No employee may approve their own originating action.
- GOV-03 Technical independence: Valid safety and professional certifications are performed by competent authorized persons; neither a GM nor a financial waiver can substitute for a mandatory technical/statutory clearance.
- GOV-04 Evidence: Worker, junior designer, client, storekeeper, inspector, supplier and auditor original submissions are preserved with authorship, timestamp, linked versions and tamper-evident history. Review adds new records; it does not overwrite originals.
- GOV-05 Segregation: Requests, verification, approval, payment, receipt and audit must be appropriately independent. Where team size makes perfect separation impractical, a documented compensating review by an independent qualified officer is mandatory.
- GOV-06 Client access: Client portal content is limited to that client/project and expressly published revision; source design files, worker complaints, confidential pay, internal contracts and audit cases remain restricted.

- GOV-07 Controls: Accounts verifies funds/beneficiaries and books actual transactions; QS values cost and forecasts; Contracts verifies commercial entitlements; Compliance oversees obligations; QA checks systemic quality; independent audit observes, investigates and independently verifies closure.
- GOV-08 Exceptions and continuity: Documented financial/administrative exceptions may be decided by the authorized delegated manager or GM; there is no exception to mandatory law, safety, professional status or an already-required client acceptance. GM absence invokes only preapproved and HR-activated acting authority.

3. Updated operating workflow and stage gates

Start

Signed construction scope permits planning only; Compliance, Accounts, site technical/safety readiness and delegated GM mobilization release unlock physical work.

Original worker reports

Every Mathmag skilled worker and subcontractor worker uses a personal account; original daily evidence is independently visible to PM and management immediately.

Materials

PO -> accepted GRN -> invoice three-way match; stock custody/issue/returns/transfers remain traceable and recipient verified.

Equipment

Unique asset ID, authorized operator competency, daily pre-use checks, fuel record, hours/cost, rented off-hire controls and restricted unsafe equipment.

Work quality

Independent stage hold-point certificates, defect logs and safety alerts; technical clearance is mandatory before unsafe or covered work proceeds.

Commercial

PM progress, QS valuations/forecast, Accounts reconciliation and management-approved client variations, subcontract payments and lawful suspension.

Closeout

Practical completion, technical approval, management-authorized handover, client response, final account and warranty admin closeout remain separate states.

4. Module-specific accountability integration

CONTROL AREA	REQUIRED V3 BEHAVIOR
Procurement authorizations	PM requests; Procurement sources; QS checks costs; Accounts checks funding and payee; qualified safety reviews technical; delegated management approves financial commitments.
Site daily reports	Original worker/driver/operator evidence independently preserved; Supervisor independently verifies; PM publishes selected client content only.
Fuel and equipment	Digital measured issues to asset/project, unique operator, logs hours; Stores+Accounts stocks, QS/PM consumption variance, independent audits; serious defect blocks equipment.
Material loss	PM and QS establish causation/cost; major/suspicious loss audit escalation; lawful employer/contractor liability only after fair due process; no automatic deductions.
Supplier advances and default	Contractual approval, QS risk/value, Accounts exposure; track offset, due notice and remedy; impairment separate from legally waiving debt.
Worker payroll	Worker reports/attendance cannot directly authorize payment; PM verifies, Accounts calculates, delegated management authorizes and Accounts disburses under separate controls.
Safety stop work	Any competent authorized site/safety role isolates immediate danger; targeted hold remains until required competent clearance and management release if serious.
Handover/warranty	PM, QS, Accounts, Contracts and professionals each verify own component; management authorizes; client acceptance does not silently waive remaining obligations.

5. Conflict resolution register

OLDER / AMBIGUOUS WORDING	V3 INTERPRETATION
“Management” every PO/repair/payroll	Decompose into delegated per-category limits, GM exception, finance checks and independent auditor visibility; owner not automatic signer.
PM “authorizes stock adjustments”	PM may authorize permissible routine adjustment following independent evidence, QS value and limits; significant/suspect cases receive higher delegated manager and audit.
“Safety and professionalism” approval	Qualified technical safety release separate from GM spend approval and repair execution.
Module 6 “last of six”	Module 7 provides constant HR, Accounts, QA/Compliance and protected independent auditing; warranty and post-handover still within Module 6.

6. Developer data extensions / shared links

Maintain one stable ProjectId across all modules; preserve exact FileVersionId and client publication authorization, verify role scope server-side and append actor/timestamp/decision history for every state transition.

- ConstructionEngagement/Mobilization
- WorkerTask/OriginalWorkReport
- Attendance/Adjustment
- Supplier/Quote/PO

- GRN/StockMovement
- EquipmentAsset/OperatorPermit
- EquipmentPreuseInspection/SafetyRestriction
- FuelIssue/MeterReading
- Subcontract/Valuation/Retention
- ProjectBudget/CostForecast
- Variation/PaymentClaim
- Handover/WarrantyClaim
- AuditEvent

7. Mandatory integration acceptance tests

AT-06-01: Original worker report reaches management before Supervisor verifies and cannot be hidden.

AT-06-02: Unverified supplier bank-change places affected payment on hold.

AT-06-03: No invoice payment with unmatched accepted GRN/approved PO unless explicitly authorized legitimate exception.

AT-06-04: No qualified operator or critical pre-use failure blocks restricted equipment.

AT-06-05: Fuel issue links to asset, project and physically acknowledged quantity.

AT-06-06: Unsafe construction stage cannot proceed on financial override.

AT-06-07: Client portal displays only individually approved progress items.

AT-06-08: Warranty case expiry cannot automatically delete or close unresolved claim.

8. Required developer references

Read this PDF with 00 Master Architecture, Module 07 Central Administration and 08 Integration & Testing. Additional authoritative user decisions for Modules 1-6 prevail over older illustrative implementation examples where the later decision is more specific. Regulatory obligations must be validated with qualified advisors and current official sources before implementation.

MATHMAG BUILDERS LIMITED

MANAGEMENT SYSTEM

MODULE 06

CONSTRUCTION & POST-HANDOVER

Workforce, inventory, quality, safety, finances, subcontractors, handover and warranties

Plan better, build better.

Organization	Mathmag Builders Limited
Purpose	Revised developer-ready functional specification
Revision	2.0 | 24 October 2026
Approval basis	Confirmed business choices + clearly identified technical implementation recommendations
System scope	Module 06 of 06, on one persistent client/project platform

IMPLEMENTATION RULE

This revision replaces inconsistent stage assumptions in earlier drafts. A confirmed business decision is binding; illustrative fields, example APIs and implementation details must be validated in design review.

01 Scope, execution status and controlling principles

Comprehensive construction operations, project controls, workforce transparency, subcontracting, client reporting, completion and post-handover warranty management.

Conditional activation	Executed construction/supervision/labour contract opens only relevant Module 6 services. Planning-only access may open while physical works remain locked.
Project Manager	Management directly appoints the Project Manager, who owns integrated project execution oversight and operational verification.
Management approval	Management controls contractor appointments, repair/contract commitments, payroll payments, major programme changes, claims, handover and warranty administrative close-out.
Independent employee reporting	Every directly hired AND subcontractor skilled worker has an account and uploads original daily photos/videos/descriptions; management sees originals before supervisor approval.
Supervisor limited verification	Supervisor independently checks reported work but may never edit, hide, delete, delay or suppress original worker evidence.
Selective client portal	Project Manager independently filters client-visible construction content. No automatic publication of all worker uploads.
Continuous audit	Link every action to actor, project, task, asset/claim/contract, timestamp and preserved revision history.

MANDATORY CONTROL

Construction mobilization is blocked until applicable permits/registration, financial and safety readiness gates are clear and management has authorized physical start.

02 Stage entry and preconstruction readiness

01	Contract Execution	Module 5 stores an executed agreement and management-appointed Project Manager.
02	Preconstruction Plan	Project Manager prepares work programme, resource needs, baseline budget, labour assignments and site readiness checklist.
03	Early Procurement Request	Any pre-mobilization purchase is itemized and individually management-authorized; Procurement handles approved purchase order.
04	Compliance Review	Verify relevant County, NEMA, NCA or other required obligations; conditional Module 4 close-out is insufficient by itself.
05	Accounts Gate	Verify contract-specific initial construction milestone or documented eligible management financial exception.
06	Project Manager Recommendation	Document manpower, site controls, access, required professionals, safety and materials readiness.
07	Management Mobilization Release	Management alone authorizes physical construction and records effective start and conditions.

03 Role catalogue and separation of duties

Management	Appoints PM; authorizes financial commitments, appointments, major schedule/scope changes, payroll, subcontract disbursements, handover and final warranty administrative close-out.
Project Manager	Owns programme, requests procurement, verifies store records and progress, monitors workers/quality/safety, selects client media, assesses claims, submits recommendations.
Site Supervisor	Assigns daily worker tasks from PM plan, changes them with reasons, verifies worker work/attendance/PPE, reports site delays/defects/incidents; cannot suppress originals.
Skilled Worker (Mathmag or subcontractor)	Personal login, check-in/out, daily task evidence, independent incident/PPE/equipment reporting, acknowledgments, correction and dispute requests.
Storekeeper	Receives/checks goods and transfer receipts, records stock issues/returns, PPE, asset handoffs, conditions and returns.
Procurement Officer	Seeks/comparisons quotations, arranges approved material orders, subcontract bids, equipment service providers and authorized purchases.
Accounts	Verifies receipts, prepares payroll and client/subcontractor payment schedules, disburses only approved transactions, maintains ledgers.
QS	Assesses valuations, change costs, subcontract quantities, final account and relevant retention sums.
Contracts Officer	Prepares subcontract agreements, variation orders and warranty-term interpretations.
Authorized Professionals	Own technical inspections, certifications and statutory clearances where required.
Client	Receives PM-authorized progress media, contractual billing, formal variation approvals, handover and warranty case portal access.

04 Procurement requisitions and approved orders

01	Project Manager	Creates resource/material requisition against activity, planned quantity, available stock and required date.
02	Procurement Officer	Obtains and compares quotations, documents suppliers, price, lead times and specifications.
03	Management	Authorizes specific vendor/amount/purchase order; extra restrictions for early procurement before mobilization.
04	Procurement Officer	Places approved purchase order and tracks shipment/partial deliveries.
05	Accounts	Processes authorized payments and retains financial evidence.
06	Storekeeper / Supervisor	Checks physical delivery; Project Manager verifies record before final stock acceptance.

MANDATORY CONTROL

The Project Manager cannot create a financial commitment merely by requesting supplies. Procurement places orders only after applicable management authorization.

05 Material delivery receiving and inspection

• Storekeeper OR Site Supervisor physically receives material deliveries and checks purchase order, delivery note, quantities, item specifications, damage and condition.

• Original delivery documents, timestamp, receiver identity, photos and any shortages are captured; receiving personnel cannot inflate actual quantity to match the order.

• Project Manager verifies receiving records and cross-checks material ordered, delivered and accepted.

• Part-delivery and rejection states remain traceable. Notify Procurement Officer of damaged, incorrect or incomplete supply.

• The system separately tracks ordered, dispatched, physically received, accepted, rejected and outstanding quantities; payments use only approved transaction conditions.

06 Material stock issuance and use

Site Supervisor	Requests project/activity-specific materials and quantities against available stock and planned work.
Storekeeper	Checks availability, issues actual quantity, records recipient, date, item, stock location and remaining balance.
Project Manager	Monitors material usage versus budget, approved material schedules, waste and productivity.
Stock record	Distinguish requested, issued, returned, transferred, consumed and adjusted quantities.
Worker activity	Link issued stock to authorized activities and daily site reporting; do not assume every issued unit was physically consumed.

07 Material wastage, discrepancies and stock adjustments

01	Issue Detected	Site Supervisor or Storekeeper reports missing, damaged, wasted or unexplained items with evidence.
02	Project Manager Investigates	Compares receiving, issuance, stock movement and actual work records; documents findings.
03	Adjustment Approved	Project Manager authorizes justified stock adjustment and quantity/cost effect.
04	Significant Loss Escalated	Management receives an alert under configured thresholds and may intervene.
05	Inventory Updated	System posts a separate adjustment transaction with original record untouched.

- Management configures significant-loss thresholds; do not invent permanent fixed values.
- No worker or supplier is automatically blamed solely due to a stock variance.

08 Inter-store transfers and future central warehouse

01	Project Manager Requests	Specifies sending store, project destination, materials, required quantities and reason.
02	Sending Storekeeper Dispatches	Records release, batch, driver/vehicle where relevant, transit status and actual quantities.
03	System Tracks Transit	Keeps in-transit inventory separate from received project stock.
04	Receiving Storekeeper / Supervisor	Checks actual condition and quantities on delivery, accepts or records shortage/damage.
05	Project Manager Monitors	Reconciles movement records and investigates discrepancies; all movements remain auditable.

MANDATORY CONTROL

Dispatched does not mean received. A receiving site balance must not increase solely on the sending storekeeper marking dispatch complete.

09 Individual worker onboarding and reporting channels

• Every directly engaged skilled worker has a unique personal account at onboarding; skilled workers employed by appointed subcontractors also receive unique accounts linked to their subcontractor employer and project.

• Worker profile includes contact, trade/competence, permitted projects, employment relationship, supervisor/PM, safety induction/PPE, assigned tasks, pay arrangement only if directly employed, and activation dates.

• Original daily work report captures assigned task, self-reported quantities, description, date/time and photos/videos; immediately visible directly to PM and management.

• A worker submission is not the same as verified work. Supervisors append independent verification status (confirmed, partial, disputed or pending) and reasons without editing or suppressing the submission.

• Mobile-first, practical low-bandwidth media uploads with draft/offline queuing, compression and confirmation of successful sync are recommended.

• Worker may contest verification through portal. PM investigates; management sees disputes independently and may intervene with a reasoned logged decision.

• Subcontractor representative has a restricted dashboard of their authorized team and project; individual worker accounts do not create an automatic direct Mathmag payroll/employment relationship.

MANDATORY CONTROL

There must be no "supervisor approval before management visibility" queue. Worker media and descriptions enter the central record immediately and remain unaltered.

10 Daily construction task planning, assignment and changes

01	Project Manager Plans	Establishes approved work programme, daily/weekly activity targets and priority order.
02	Site Supervisor Assigns	Allocates worker-specific task, location, quantity, expected finish and instructions.
03	Worker Receives	Assignment is immediately available in the individual portal; PM and management can also see it.
04	Conditions Change	Supervisor revises routine assignment/target with mandatory reason and immediate PM/management visibility.
05	Worker Reports	Uploads actual task progress and evidence against the applicable version of task.
06	Verification and Review	Supervisor inspects work; PM compares original/revised target, worker claim and verified result.

MANDATORY CONTROL

Changes materially affecting budget, approved project programme or contractual scope must be escalated for Project Manager/management approval under the relevant change-control policy. All original task targets are retained.

11 Worker daily reports, supervisor verification and site-level reports

Individual worker original	Photo/video + short description, task/worker/project/date and self-reported quantity; immediately visible to PM/management.
Supervisor work verification	Records actual observed quantity/quality and status; never edits worker original.
Supervisor daily site report	Separate daily overview: activities, work quantities, attendance, materials consumed, equipment, weather, challenges, safety, quality and next actions.
Project Manager review	Validates site report; approves or returns for correction; compares totals against original worker evidence.
Performance evidence	Stores verified output, deadlines, rework and disputes with attribution and contextual causes.
Client publication	PM selects exact reviewed photos/videos/report extracts for client portal; no blanket publishing of uploads.

12 Client progress updates and media selection

• Project Manager reviews approved daily site reports and worker-generated media, then authorizes specific photos/videos/descriptions for that client project.

• Client sees only approved media and construction summaries, not original work disputes, confidential employee performance, raw financial records or internal safety investigation details by default.

• Publication is an explicit item-level grant with PM ID, selected version, date, project and revocation/supersession history; originals remain internal.

• Construction update frequency must follow the project-specific contract or management-approved arrangement; earlier choices about frequency were recommendations, not a confirmed uniform daily/weekly policy.

• Client can view construction progress independent from invoices and warranty case statuses, each governed by distinct publication and visibility controls.

13 Individual and team performance evaluation

Skilled worker	Assigned vs verified quantity, time, reporting consistency, workmanship, defects/rework attributable after investigation, contextual constraints.
Site Supervisor	Timely verification, quality/site report accuracy, task planning, dispute patterns, hazard response, delays and inspection oversight.
Project Manager	Programme adherence, risk response, accurate verification, cost/stock oversight, closure timeliness, client communication and team safety.
Storekeeper / Procurement	Inventory accuracy, traceability, compliant issue/receiving, lead time, authorized orders and supplier quality.
QS / Technical Staff	Technical completeness, quantification accuracy, review speed, rework and approved work quality.
Subcontractor team	Verified progress, quality, schedule, safety and defect-response performance at contract/team level.

MANDATORY CONTROL

Unresolved disputes are not automatic performance penalties. Compare employees on role-relevant denominators and consider missing materials, unsafe conditions, weather, client changes and technical errors outside their control.

14 Attendance, corrections and employee pay arrangements

Independent attendance

• Workers check in/out using their own portal. Supervisor separately verifies on-site presence/arrival/departure; PM and management compare both sets.

• Original worker timestamps must never be replaced. Missing entry corrections are requested by worker with reason, checked by supervisor and approved or rejected by PM; post an adjustment record.

• Keep check-in/out date, assigned site, verifier, attendance exceptions, evidence and audit timestamps; align payroll to relevant verified attendance.

Flexible wages / payment terms

• Support daily wage, weekly wage, monthly salary, piecework/quantities and fixed task agreements at individual worker level.

• PM verifies eligible attendance/work and recommends pay; Accounts calculates/validates payroll; management finally authorizes; Accounts disburses and records transactions.

• Worker portal displays own wage terms, payable/approved amounts, statements, prior payments and permitted deductions, never other employees financial details.

• If part of work is contested, segregate disputed quantities from undisputed amounts and apply the actual contract and applicable law.

15 Worker pay advances and recovery

01	Worker Requests	Individual portal form captures requested amount, reason and date.
02	Project Manager Recommends	Checks worker engagement, related project and operational context.
03	Accounts Verifies	Reviews earned amounts, prior advances, eligible recovery terms and limits.
04	Management Authorizes	Approves, rejects or amends amount and lawful recovery terms.
05	Accounts Disburses / Recovers	Records paid advance and agreed, legally permitted recovery against ledger.
06	Worker Views Ledger	Displays outstanding advance, approved recoveries, dates and history.

MANDATORY CONTROL

Do not automatically deduct money merely because an advance was requested. Recovery must follow documented agreement and applicable law.

16 Quality defects and mandatory construction hold points

Construction defect management

• Supervisor creates defect with location, condition, severity, evidence, assigned corrective task and deadline.

• Individual workers independently upload repair evidence; Supervisor checks correction; PM verifies and authorizes closure.

• Management can inspect originals, repeat defects, rejected corrections and reasons; no silent deletion or attribution of fault.

Mandatory stage inspections

• Supervisor uses checklist/photos to request inspection before concreting, covering up services or moving beyond configured critical stage.

• PM verifies readiness and authorizes progression only after every required independent professional/statutory technical clearance is recorded.

• Failed or pending check leaves stage on hold. Original report, professional certificate and PM release are separate audited decisions.

• Applicable checks may include foundations, reinforcement, DPM, suspended slabs, concealed services, roof water proofing and trade-specific tests.

17 Delay reports, recovery and construction programme revisions

• Supervisor logs delay cause, event time, work packages, evidence, impact and mitigation on the day it occurs.

• PM assesses time, critical path, procurement, labour, cost and client consequences; may direct routine recovery within approved contract/budget/programme.

• Any major completion-date, budget, scope or contractual commitment change requires management authorization and client notice/acceptance where contractually required.

• Maintain original baseline programme, every approved revised programme, change rationale, dated decisions and activity dependency links.

• Delays unrelated to worker performance must not automatically penalize individual productivity indicators.

18 Construction variations and additional works

01	Request Registered	Client or authorized project actor proposes altered/additional work with origin and description.
02	Project Manager Assesses	Reviews physical scope, schedule, feasibility and operational impacts.
03	Relevant Professionals Review	Check design implications, technical safety, revisions and permit needs.
04	QS Costs	Prepares variation prices, allowances, credits and resource/time impacts.
05	Contracts Officer Documents	Creates controlled variation order and amended conditions.
06	Management Authorizes Offer	Approves scope, pricing, payment arrangement and client-facing issue.
07	Client Formally Accepts	Signs or accepts exact terms before variation is executable.
08	Finance/Compliance Checks	Accounts verifies any contractually due payment gate; mandatory professional/statutory prerequisites remain enforced.
09	Project Manager Releases Execution	Adds variation tasks to approved work plan without overwriting original contract.

MANDATORY CONTROL

A Site Supervisor instruction alone cannot create chargeable extra works; an approved variation requires technical review, QS assessment as needed, management offer approval and formal client acceptance.

19 Flexible variation payments and client construction payment claims

Variation advance / milestones	Contracts Officer defines variation-specific terms; management approves; client accepts; Accounts verifies amounts due at the relevant execution point.
Construction milestone claim	PM validates milestone evidence; management authorizes claim; Accounts issues invoice and verifies receipt.
QS-measured progress	PM confirms physical progress; QS values work where contract requires; management approves claim; Accounts invoices and records receipts.
Scheduled instalment	Accounts issues contractual invoice according to agreed dates after required checks and authorization.
Hybrid contract terms	May combine schedule, milestone and valuations; configure per signed contract rather than one universal method.
Account ledger	Track contract sum, signed variations, claims, certified amounts, invoices, verified receipts, balances and adjustments separately.

20 Overdue client payments and contractual suspension

01	Accounts Flags Overdue	Records amount, due date, missed milestone and client communication/reminders.
02	Project Manager Assesses	Evaluates exposure to site resources, labour, procurement, programme, safety and contract.
03	Management Decides	May authorize continued works, agreed payment plan or lawful suspension after applicable notice and safety requirements.
04	Project Holds Updated	System records decision, scope, required notices, demobilization/safety precautions and authorized effective date.
05	Accounts Verifies Recovery	Records receipts and remaining obligations; management authorizes resumption where necessary.

MANDATORY CONTROL

An overdue invoice must not automatically suspend construction. Management makes the decision under the signed contract, applicable law and site-protection requirements.

21 Safety hazards, incident reporting and work resumption

Independent hazard and incident channel

• All workers can independently report hazards, near misses, incidents and injuries with photos/videos; original report directly reaches PM and management without supervisor approval.

• Supervisor takes immediate protective actions, including halting unsafe activity and requesting emergency help. Critical cases generate immediate management escalation.

• PM investigates, records contributing factors and tracks corrective actions, with competent safety/technical personnel and statutory notifications where required.

Release after hazard correction

• Supervisor independently checks corrective actions and records evidence; PM may authorize routine resumption.

• Serious events require management authorization and all applicable competent professional or regulator clearance before affected activity resumes.

• Retain safety hold, inspection outcome, clearance certificates, manager decision and date. Worker reporting must not trigger retaliation.

22 Safety induction, toolbox talks and PPE

Induction and talks

• Supervisor conducts site induction and toolbox briefings; logs topic, hazards, instructions, trainer, date and attendees.

• Every worker independently acknowledges attendance through own portal, immediately visible to PM/management; acknowledgments do not substitute for competency checks.

• PM verifies compliance; mandatory inductions, qualifications and briefing conditions gate hazardous assignments.

PPE issuance

• Storekeeper issues role-appropriate PPE to named worker and records size/type/condition/date; worker independently acknowledges receipt.

• Supervisor verifies availability, fit, condition and use; PM monitors compliance and management sees all records.

PPE replacement

• Worker independently requests replacement of lost/damaged/unsuitable/worn PPE; supervisor checks; Storekeeper promptly provides replacement and updates stock.

• Worker acknowledges new issue. Essential PPE must never be delayed as punishment; affected hazardous activity is held or safe alternative task offered.

23 Tools and equipment custody, maintenance and repair

Asset custody

• Supervisor requests tools for assigned work; Storekeeper issues against uniquely identified item and responsible worker/operator.

• Worker independently acknowledges receipt and return through portal; Storekeeper physically verifies returned condition; PM tracks location and overdue assets.

• Powered/restricted equipment is allocated only to authorized competent personnel.

Breakdowns and repairs

• Worker independently reports fault; Supervisor or Storekeeper checks asset; unsafe machine is marked Out of Service.

• PM coordinates diagnostic/repair; management approves repair expenditure; Procurement arranges approved supplier/service; Accounts pays according to approved route.

• Equipment cannot be released for reuse until documented inspection and any required safety/technical testing.

Preventive maintenance

• System tracks service based on dates, logged operating hours and manufacturer schedules and issues upcoming/due/overdue reminders.

• PM coordinates; Procurement arranges management-authorized servicing; retain service history, costs, operating hours and release-to-service clearance.

24 Subcontractor appointment, worker accounts and oversight

01	Project Manager Recommends	Identifies specialist scope and technically appropriate candidate providers.
02	Procurement Quotes	Obtains and compares quotations, credentials, lead times and commercial details.
03	Contracts Officer Drafts	Prepares subcontract, scope, pricing, milestone, warranty, safety and retention terms.
04	Management Appoints	Authorizes binding subcontract commitment and contractor engagement.
05	Subcontractor Onboards	Representative portal and personal accounts for each skilled subcontract worker are created and scoped to project.
06	Project Manager Supervises	Monitors performance, individual worker evidence, defects, safety, schedule and contract compliance.

MANDATORY CONTROL

Subcontractor worker reporting rights do not automatically create a direct Mathmag wage obligation; employee and subcontractor contractual relationships remain separate.

25 Subcontractor claims, retention and final payment

• Representative submits subcontractor payment application through portal against agreed contract, progress and attachments.

• PM verifies physical progress, quality and site evidence; QS values quantities where required by subcontract; Accounts checks prior claims, variations, deductions and agreed commercial terms.

• Management authorizes final payment schedule; Accounts disburses and records the transaction. Worker daily reports do not independently authorize subcontractor payments.

• Retention is flexible per subcontract: none, fixed amount or percentage, cap, release milestones, defect period and conditions where agreed.

• PM verifies completion/defects; QS assesses final account if required; Accounts reconciles held/released retention; management explicitly authorizes each release.

• Passing retention expiry does not automatically release money unless the relevant payment conditions and legally applicable obligations have been satisfied.

26 Subcontractor completion, inspections and handover

01	Subcontractor Requests Handover	Representative submits completion package and related worker evidence.
02	Site Supervisor Inspects	Records physical inspection, snag list, quantity and quality observations.
03	Workers Correct Snags	Individual subcontract workers upload independent correction evidence, linked to defects.
04	Project Manager Verifies	Checks scope completion, defect closure and any permitted pending obligations.
05	Required Professional Clearance	Independent authorized professionals provide specialist/technical certificates where necessary.
06	PM Records Recommendation	Issues completion recommendation with evidence and conditions, visible to management.
07	Final Account Proceeds	Completion supports valuation/retention review but does not itself authorize Accounts to pay.

27 Main construction completion and client handover

01	PM Prepares Completion	Organizes final inspections, final works documentation and defect close-out.
02	Technical Professionals Sign	Provide applicable completion, system-testing or statutory certificates.
03	QS and Accounts Reconcile	Check approved variations, contract value, final account, invoices, balances and retention terms.
04	PM Compiles Pack	Applicable as-built documents, manuals, warranties, certificates, photos and handover record.
05	Management Authorizes	Reviews technical, financial and contractual readiness and permits formal client handover.
06	Client Accepts	Signs at physical meeting or accepts through client portal; record any expressly agreed minor remaining items.
07	After-Handover Starts	Warranty/defects-liability clocks and continuing obligations tracked under actual contract trigger dates.

MANDATORY CONTROL

Physical completion, client handover, financial close-out and end of defects-liability obligations are distinct. Client sign-off must not silently waive unresolved or statutory responsibilities.

28 Post-handover defect claims, coverage and costs

01	Client Claims	Client opens claim via portal, attaches descriptions/photos/videos and marks urgency.
02	Project Manager Investigates	Records inspection, cause, repair evidence and immediate safety response if warranted.
03	Professionals Assess	Specialist/structural/safety defects receive independent technical review as needed.
04	Contracts Officer Reviews	Checks original construction agreement, warranty/defects liability, supplier/subcontractor responsibilities.
05	QS Costs Where Needed	Prepares repair-cost estimate if cost allocation or chargeable work requires.
06	Management Resolves	Approves covered remedy/cost allocation or offer of chargeable work.
07	Client Accepts Charges	Non-emergency chargeable correction requires client acceptance of documented scope, price and terms.
08	Worker/Subcontractor Repairs	Original corrective evidence entered through assigned portal, independently visible to management.

29 Priority response, verified closure and no-response rule

Response-time priority

• Classify claims Critical, Urgent or Routine. Management configures separate initial-response and resolution target intervals consistent with contract, severity and legal obligations.

• System timestamps claim, acknowledgment, inspection, response, resolution; escalates approaching/overdue milestones to PM and management.

• Critical safety hazards receive immediate protective or emergency action regardless of liability and ordinary business approvals.

Client confirmation and dispute

• PM verifies repairs and obtains mandatory technical clearances; client reviews evidence and explicitly confirms satisfaction or disputes findings in portal.

• Disputed claims remain under investigation with independent management oversight; client-confirmed and management-administrative closures have distinct statuses.

Client non-response

• System sends reminders at management-set intervals and logs delivery/response. PM compiles completed repair evidence and communication history.

• Management alone may authorize a reasoned administrative closure if contract and law permit. Client silence is never satisfaction or release of rights.

30 Reopening claims, expiry tracking and final warranty close-out

Reopening

• Client can request reopening of a previous claim with new evidence; PM reviews original history and investigates. Management sees all requests and can intervene in contested decisions.

• Original complaints, prior repairs and closures are retained. Reopening does not automatically prove eligibility for free work or extinguish applicable rights.

Expiry monitoring

• Maintain contract-specific warranty and defects-liability start/end dates for overall building and separately for trades, suppliers, components and subcontractors if applicable.

• System calculates due dates, sends reminders at management-defined intervals and alerts management of approaching/passed expiry and outstanding claims.

• Expiry never erases, auto-closes or prejudges unresolved/reopened claims or statutory/contractual duties.

Final administrative close-out

• Project Manager prepares a final report: expiry periods, original/reopened claims, repairs, client confirmations/disputes, administrative no-response closures, outstanding defects, clearances and continuing obligations.

• Management independently reviews and alone authorizes administrative close-out when appropriate. The system must not label it "all liabilities discharged".

• Retain evidence, employee uploads, decisions, correspondence and re-opening capability subject to applicable retention/legal policies.

31 Shared rules carried across all six modules

One lifecycle record	Maintain one project ID from confirmed site-visit booking through design, approvals, contracts, construction and post-handover. Never create a disconnected project just to open a new module.
Role separation	User role, project assignment and explicit approval authority determine every permitted action. Technical sign-off, financial validation, management approval and client acceptance are distinct records.
Evidence integrity	Preserve original submissions, attachments, timestamps, authors and immutable decision histories. Corrections create new records or versions; no silent overwriting.
Selective client access	Client accounts can access only their own projects and explicitly authorized files or updates. Internal working/source files, staff performance data and financial notes are never published by default.
Conditional service scope	Clients may purchase design only, printing only, approvals, technical supervision, labour-only building or full construction. Modules not contracted are not mandatory project steps.
Management-controlled exceptions	Any permitted exception requires identified management authorization, reason, date and retained evidence. Mandatory legal/technical clearances cannot be overridden.
Security and audit	Require authenticated accounts, least privilege, encrypted transport/storage, file scanning, access logs, project-scoped permissions, backups and recoverability.
Work reporting fairness	Worker-originated material is permanently visible to Project Manager and management separately from Site Supervisor verification; manager decisions can override erroneous assessment with reason and without alteration of originals.

32 Screens and user journeys

Project Manager Command Center	Projects, programme, financial gates, exceptions, daily reports, live delays and approval queues.
Worker Mobile Portal	Assigned task, evidence upload, check-in/out, correction/dispute, incident, PPE, tool custody, earnings and advances where applicable.
Supervisor Site Dashboard	Daily task assignments, versioned changes, individual work verification, attendance, defects, safety and stage inspections.
Central Store / Site Store	Receiving, stock balances, issue request, consumption, returns, transfers, adjustments, PPE, assets.
Procurement Workspace	Requisitions, quotes, supplier bids, purchase orders and approved repairs.
Client Construction Portal	PM-approved progress photos/videos, selected reports, milestone invoices, variations, handover and warranty support.
Management Oversight	Original worker evidence vs supervisor verification, disputes, safety alerts, PM recommendations and approval decisions.
Accounts and Payroll	Worker advances, pay recommendations, payroll approval, subcontract claims, client invoicing and receipts.
QS Valuation Desk	Variation costing, subcontract payment valuations, progress claims and final account.
Subcontractor Representative	Agreement, workers/tasks, representative claims, compliance, handover and retention ledger.
Quality / Safety Register	Construction hold points, defects, safety incidents, technical clearances and release conditions.
Asset Management	Equipment custody, operating hours, defects, repair authorizations and preventive service alerts.
Warranty and Close-Out	Client claims, severity timers, corrective works, re-openings, expiries, final management close-out.

33 Core data objects and key relationships

Logical entities for developer schema design. Names below are examples rather than prescribed database table names.

ConstructionEngagement / Mobilization	Executed Module 5 service scope, appointed PM, readiness gates, permit IDs, management approval.

WorkPlan / Activity / WorkerTask	Baseline/current programme, task versions, assigned workers, quantities, deadlines, change reason.
WorkerProfile / SubcontractorEmployeeLink	Trade, employing entity, verified permissions, project allocation, induction, PPE, payment relationship.
WorkerWorkReport / WorkerMedia	Original worker submission, hash, timestamp, task/project, self-reported quantity and independently stored evidence.
SupervisorVerification / WorkerDispute	Separate observed work, status, findings, PM resolution and management intervention.
AttendanceEvent / AttendanceAdjustment	Original check-in/out record; supervisor verification; worker-initiated correction and PM authorization.
MaterialStock / StockMovement	Item/batch, store, receipt, issue, return, consumption, transfer, variance and adjustment reference.
PpeIssue / PpeReplacement / Acknowledgment	Worker-specific equipment, issuance, inspection, return/replacement and direct worker confirmation.
EquipmentAsset / Custody / Maintenance	Asset ID, operator, usage, faults, repair approvals, service schedule and inspection release.
ConstructionDefect / StageInspection / SafetyIncident	Severity, photo/video evidence, technical clearance, work holds, corrective action and audited closure.
Subcontract / Claim / RetentionLedger	Representative account, signed terms, valuations, staged payment, defect conditions and authorized releases.
VariationOrder / PaymentClaim / ClientLedger	Technical/QS/contract review, client acceptance, claim, invoice, payment and financing gate.
HandoverCertificate / WarrantyTerm / WarrantyClaim	Client acceptance, coverage, expiry, cases, reopening history, admin versus satisfactory close-out.
PublicationGrant / AuditEvent / Notification	File-specific PM publication, actor, state changes, immutable audit and channels.

34 Illustrative service and event contracts

Suggested API/service boundaries for planning. Authentication, validation, permission checks, idempotency and audit metadata apply to all write operations.

Worker evidence	POST /projects/{id}/worker-reports with worker task, media and immutable original.
Supervisor check / dispute	POST /worker-reports/{id}/verifications; POST /verifications/{id}/disputes.

Daily tasks/changes	POST /projects/{id}/worker-tasks; POST /worker-tasks/{id}/revisions.
Attendance and adjustment	POST /workers/{id}/attendance-events; POST /attendance/{id}/correction-requests.
PM client media selection	POST /projects/{id}/client-media-publications with exact media IDs.
Stock and assets	POST /stores/{id}/stock-issues; POST /stock-transfers; POST /assets/{id}/maintenance-orders.
Safety and quality gates	POST /projects/{id}/incidents; POST /stage-inspections/{id}/pm-clearance.
Payroll approvals	POST /projects/{id}/pay-recommendations; POST /payment-schedules/{id}/management-authorization.
Subcontract claims	POST /subcontracts/{id}/claims; POST /claims/{id}/qs-valuation.
Construction variations	POST /projects/{id}/variation-requests; POST /variations/{id}/client-acceptance.
Warranty cases	POST /projects/{id}/warranty-claims; POST /warranty-claims/{id}/reopen.
Events	WorkerReportSubmitted, SupervisorVerificationRecorded, WorkDisputed, SafetyHoldApplied, ProfessionalClearanceRecorded, ClientMediaPublished, PaymentAuthorized, WarrantyClaimExpiredPeriodWithOpenCase.

35 Mandatory notification and automation matrix

Original worker report submitted	Project Manager and management: immediately available; Site Supervisor gets verification task.
Supervisor reports conflict	Worker, Project Manager and management can view reasons; PM receives decision queue.
Safety critical incident	Immediate alert to management/PM; site protective action proceeds immediately.
Stock shortage / significant adjustment	Procurement/PM alerts; management escalation on configured significant threshold.
Task revised	Worker notified; PM and management receive original-versus-revised visibility.
Payroll/contractor payment awaiting approval	Management decision queue; Accounts cannot disburse until authorized.
Stage inspection hold / clearance	Assigned responsible inspector/professional/PM notified; next activity remains blocked.
Critical equipment service overdue	Storekeeper/PM and management; restrict unsafe equipment.
Warranty response due/overdue	Assigned PM plus management escalation under configured severity target.
Warranty approaching expiry	Project Manager, management and authorized client reminders on selected configured schedule.

36 User acceptance tests and release criteria

• TEST 01 - Permit Module 6 planning-only access after contract execution, but deny on-site task start until approved compliance, financial, readiness and management mobilization gates.

• TEST 02 - Independent skilled worker report appears in PM and management records before Site Supervisor action; supervisor cannot change/delete evidence.

• TEST 03 - Subcontractor worker has distinct personal portal and management-visible originals; no automatic direct Mathmag payroll account created.

• TEST 04 - Project Manager selects individual verified media to publish; client cannot see unselected worker uploads or internal employee reviews.

• TEST 05 - Site Supervisor may revise routine daily tasks only with reason and audit; original targets retained; major programme change escalates.

• TEST 06 - Worker check-in/out timestamps are immutable; PM-approved corrections create separate adjustment and original remains visible.

• TEST 07 - Inventory dispatch does not increase receiving site balance until actual acceptance; partial damaged deliveries handled.

• TEST 08 - Stock discrepancies require PM adjustment approval and management escalation when threshold exceeded.

• TEST 09 - Management alone approves payroll, significant repair commitments, subcontract appointment/payment, major programme change and final handover.

• TEST 10 - Hold concrete placement on pending required stage inspection/professional clearance and prevent bypass.

• TEST 11 - Unsafe equipment and serious incident holds prevent use/resumption until all applicable technical/statutory and management approvals.

• TEST 12 - Variation without management-approved price and formal client acceptance cannot become an executable extra task.

• TEST 13 - Overdue client invoice alerts but cannot automatically suspend construction without lawful management decision.

• TEST 14 - Subcontract retention remains contract-specific; no universal percentage or auto-release of funds.

• TEST 15 - Warranty case with no client response may be administratively closed only by management and never marked client-satisfied.

• TEST 16 - Reopen warranty cases while preserving originals; warranty expiry alone never erases or closes claims.

• TEST 17 - Final warranty administrative close-out requires Project Manager final report and independent management authorization.

37 Phased implementation priorities and developer handover

Wave 1 - Core field evidence	User/roles, project/site, individual worker and subcontractor portals, original media storage, supervisor verification, PM/management side-by-side view, daily site reporting.
Wave 2 - Operational stores and people	Procurement requests, goods receiving, inventory issue/transfer, attendance, worker tasks/changes, PPE, equipment custody and payroll approvals.
Wave 3 - Controlled execution	Stage inspections, hazards, quality defects, programme changes, subcontract appointment/claims, client publication and financial controls.
Wave 4 - Commercial completion	Variation orders, client progress invoices, handover, warranty claims, response timers, expiry and management close-out.
Cross-cutting	Responsive mobile portal, secure uploads, notification queue, managed object permissions, audit logs, backups, test data migration and monitoring.

MANDATORY CONTROL

Waves are an implementation recommendation, not confirmed management policy. Business rules above take precedence over convenient developer shortcuts.